FRDM
Development Platform

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Design without bounds with FRDM

*Optional

One platform, endless possibilities

Software & tools
Comprehensive software and tools for seamless prototyping and rapid development

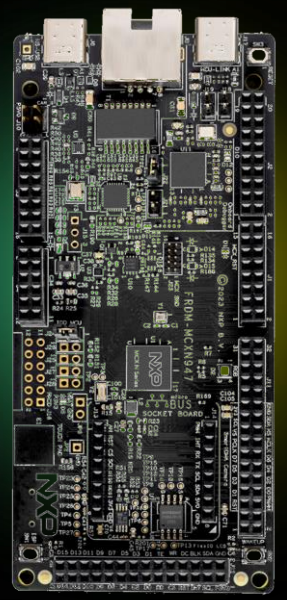
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Modular Hardware
Scale your project with modular, quick-start FRDM and expansion boards

+

Application code
Leverage our application code hub or GoPoint to access 180+ code snippets and demos

General purpose and Wireless MCUs



Flexible and scalable hardware
Arduino
FRDM
mikroBUS

Price range:
$10 - $35

i.MX Application Processors

Linux Platform with easy setup
Wireless connectivity
HDMI*
40-pin EXPI

Price range:
$50 - $150



Addressing a wide range of edge applications with baseline features across the platform

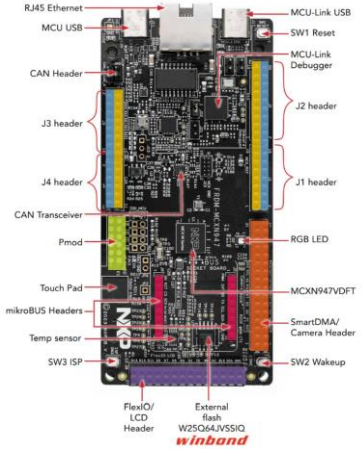
Get Started is easy

FRDM
Find it.
Get it.
Use it.

Get Started

Excellent Out-of-Box Experience Packed with helpful information and a step-by-step web guide to get you up and running quickly.

- Discover Your Board
- Install the Software
- Power Up and Run
- Build Your First Project
- Dive into More Resources

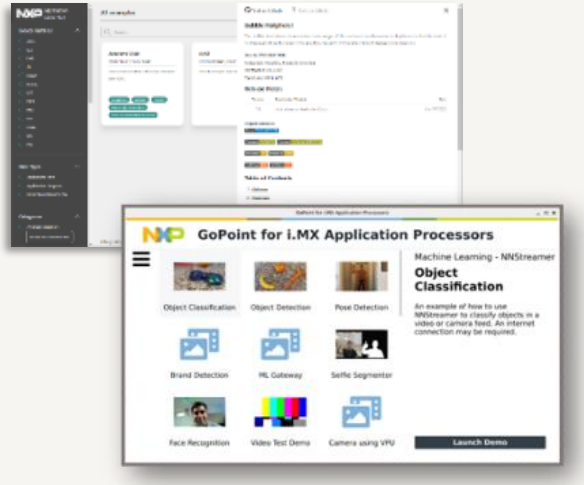


Start with our boards now:
nxp.com/FRDM-MCXN947/start
nxp.com/FRDM-MCXA153/start

Visit nxp.com/FRDM for more options and to get to know all the FRDM development boards.

Develop

Accelerate development with open-source code examples for use cases such as vision, ML, graphics, and more in Application Code Hub and GoPoint OOB demo experience in Linux BSP



Visit FRDM Community & learn

- Software and tools
- Access to training material
- Know how.. ? articles
- Open-source resources
- And more.

Get in touch

Explore FRDM

nxp.com/FRDM

Sales points

Buy Direct or through Distribution Network

Support

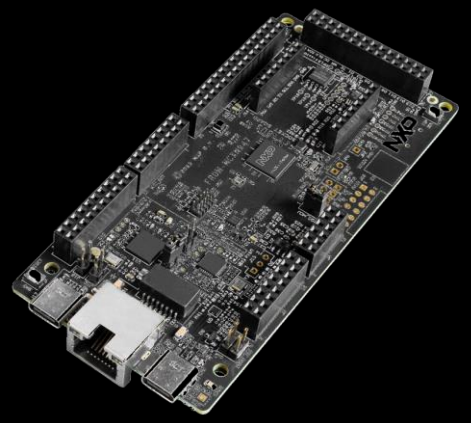
- NXP FAEs and NXP Dedicated FAEs within Distributors. Contact your sales representative.
- Web Support
  - Communities
  - Live Chat
  - Raise a ticket

Need more support?

Professional Support goes beyond our complimentary standard support, providing you with a direct link to knowledgeable NXP specialists who are dedicated to accelerating your time to market.

Cut through complexity & scale with NXP's FRDM ecosystem

FRDM Development Platform



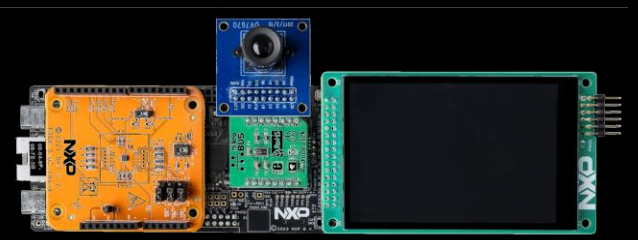
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Expansion Boards with SW support



=

Easy proof of concept



Enabling technologies with modular hardware and comprehensive software supported by SDK, GitHub or GoPoint

AI/ML	Security	Touch	Motor control	Safety	Industrial Networking
Graphics & Display	Wireless Connectivity	Voice	Vision	Power Conversion	Ultra-Low-Power

FRDM Platform addressing focus use cases

* Coming soon
**Coming later

Compact, flexible and scalable hardware design.

Product Series	FRDM Development Board	Main use case	Board Key Features
i.MX RT1180	FRDM-IMXRT1186	Industrial Networking for Factory Automation	2x Ethernet TSN, 2x Ethernet EtherCAT or TSN switch, 256 Mbit HyperRAM, 512 HyperFlash, 128 MB QSPI Flash, USB Type-C, CAN transceiver
i.MX RT700	FRDM-IMXRT700 **	Smart home	MIPI DSI/CSI, Mics, Codec, on-board Wi-Fi, FlexIO, RAM and Flash
MCX A	FRDM-MCXA153	Smart sensing node, motor control	128KB Flash, 32 KB RAM, FS USB, 1x ADC 16b
MCX A	FRDM-MCXA156	Smart sensing node, motor control, Industrial HMI	1 MB Flash, 128 KB RAM, FlexIO, CAN FD, FS USB, 2x ADC 16b
MCX A	FRDM-MCXA174	Smart sensing node, motor control	180MHz MHz Cortex-M33, 256KB Flash, FS USB, ADC, CAN-FD, FlexIO
MCX A	FRDM-MCXA266	Smart sensing node, motor control, Industrial HMI	240 MHz Cortex-M33, 1MB Flash, FS USB, SLCD, CAN-FD, Adv. security, FlexIO
MCX A	FRDM-MCXA344	Smart sensing node, motor control	180 MHz Cortex-M33, 256KB Flash, ADC, CAN-FD
MCX A	FRDM-MCXA346	Advanced Motor Control	Cortex®-M33 180MHz, 1MB Flash, 2x FlexPWM, MAU,
MCX A	FRDM-MCXA366	Smart sensing node, motor control, Industrial HMI	Arm Cortex-M33, 1MB Flash, FS USB, SLCD, CAN-FD, Adv. security, FlexIO,
MCX A	FRDM-MCXA577**	Smart Building, Factory Automation, Motor Control, Industrial Networking	Arm Cortex-M33, 2MB Flash Dual bank, HS USB, LP SPI/I2C/UART, CAN-FD, Adv. security, FlexIO, Ethernet
MCX C	FRDM-MCXC041	Sensor node, power tools	32KB Flash,2KB RAM
MCX C	FRDM-MCXC242	Sensor node, power tools	256 KB Flash, 32KB RAM, FS USB
MCX C	FRDM-MCXC444	Sensor node, power tools, essential display	256 KB Flash, 32KB RAM, FS USB, SLCD, Visible light sensor
MCX C	FRDM-MCXC162*	Smart Node, Motor Control	Arm Cortex-M23, 64KB Flash, 12KB SRAM, 16-bit ADC, DAC, OpAmpm, FlexPWM, RTC, LPUART, LPSPI, LPI2C
MCX E	FRDM-MCXE247	Motor drive, heat pump, HVAC, Power tools, BMS	Cortex®-M4F, 5/3.3V, ETH, 3x CAN FD, 2MB Flash + 8MB Ext ,
MCX E	FRDM-MCXE31B	Functional Safety for Factory Automation	Cortex®-M7 , 5/3.3V, ETH, 3xCAN FD, 4MB Flash + 8MB Ext
MCX L	FRDM-MCXL255 *	Low Power Sensing for Measurement & Metrology	Dual core Arm Cortex®-M33 & M0+ , SLCD, 1.8 - 3.3 V
MCX N	FRDM-MCXN236	HMI, Motor Control,	Cortex-M33 150 MHz, 1MB Flash, HS USB, CAN-FD, Parallel display and camera, ext Flash
MCX N	FRDM-MCXN947	TinyML	Dual Cortex-M33 150 MHz, 2MB Flash , NPU, Ethernet, HS USB, CAN-FD, Parallel display and camera, ext Flash
MCX W	FRDM-MCXW23	Portable medical deviced, smart sensing, asset tracking	Cortex-M33, 1MB Flash, small form-factor, Buck DCDC, Battery connector
MCX W	FRDM-MCXW71	Smart lighting, metering, sensor	Cortex-M33 96MHz, Multiprotocol connectivity BT LE 5.3 and 802.15.4 , USB, RF Connector, PCB Antenna
MCX W	FRDM-MCXW72	Matter Enabled Smart Home End Points	Cortex-M33, Bluetooth channel sounding support, 2MB Flash, 256 KB RAM for App, CAN, Edgelock SE
RW	FRDM-RW612	Matter Enabled Smart Home Border Router	Cortex-M33 260MHz, QSPI Flash & PSRAM on-board, flashless, Ethernet, PCB Antenna, HS USB

* Coming soon
**Coming later

FRDM Platform addressing focus use cases

Compact, flexible and scalable hardware design.

Product Series	FRDM Development Board	Main Use case	Board Features
I.MX8MPLUS	FRDM-IMX8MPLUS	Advanced Industrial vision, HMI	Arm Cortex-A53 + Cortex-M7, ML Accel 2+ TOPS, IW612, 2x Etheret, LVDS, HDMI, MIPI CSI/DSI, LPDDRA 16b 4GB, eMMC 32GB, MicroDS card
I.MX91	FRDM-IMX91	Gateway, Indutrial HMI	Arm Cortex-A55, IW610, 1GB DRAM, 8GB eMMC, SD Card, 2x Ethernet, M.2 (Key E) *rework needed, paralllel display and camera
I.MX91	FRDM-IMX91S	Gateway, Indutrial HMI, Linux optimized (light version)	Arm Cortex-A55, IW610, 512MB DRAM, 256 MB eMMC, SD Card, 1 Ethernet, paralllel display and camera
I.MX93	FRDM-IMX93	Gateway, Indutrial HMI, Industrial vision	Arm Cortex-A55, Arm Cortex-M33, IW612, 2x Ethernet, HDMI, MIPI-CSI/DSI, LPDDRA 16b 2GB, eMMC 32GB, MicroDS card
I.MX93W	FRDM-IMX93W**	Gateway, Indutrial HMI, Industrial vision	Arm Cortex-A55, Arm Cortex-M33, on-chip Wireless radio, 2x Ethernet, HDMI, MIPI-CSI/DSI, LPDDRA 16b 2GB, eMMC 32GB, MicroDS card
I.MX943	FRDM-IMX943 **	Industrial networking(PLC, TSN, EtherCAT), motor control	HS Connector, 2x GB Ethernet, M.2 Key M, M.2 Key E, DMICs, MQS, HDMI, CAN, PMIC (PF9455, PF53)
I.MX95	FRDM-IMX95 *	Advanced Industrial vision, HMI	(15x15 Package)4x ACortex-A53 + -M7, ML Accel 2+ TOPS, IW612, 2x Ethernet, LVDS, LVDS to HDMI, MIPI CSI/DSI, LPDDRA 16b 8GB, eMMC 32GB, MicroSD card, M.2 Key E, M.2 Key M (1x Ara modules) , 2x PF53, PF09,
I.MX95	FRDM-IMX95-PRO **	Generative AI & Vision for the Edge	(19x19 package)MIPI CSI/DSI, HDMI, 2x GB ENET, MicroSD Card slot, M.2 Key E, 2x M.2 Key M (2x Ara modules), 2x PF53, PF09, SE051, EEPROM, LPDRR5 16GB, eMMC 5.1 32 GB, QSPI 512 Mbit

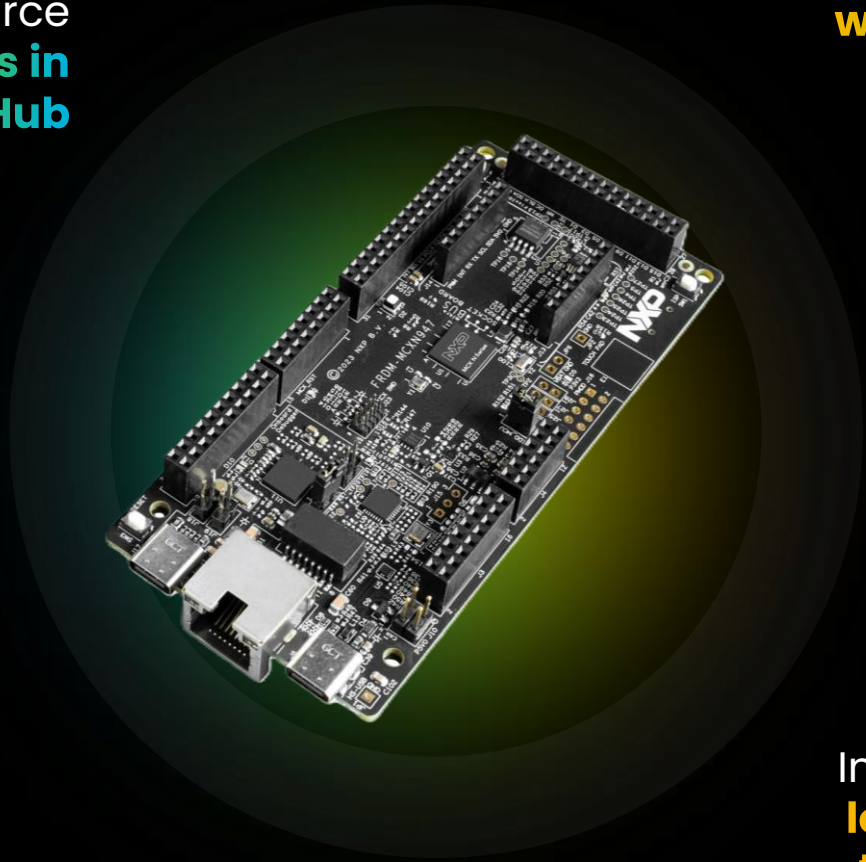
FRDM Development Ecosystem: flexible and rapid development

Join our open-source community of developers migrating to NXP

NXP & Partner open-source code **examples in Application Code Hub**

125 MCX orderable part numbers released in 2024, **on track to double in 2025**

Zephyr RTOS



Join to the **users developing with MCX now!**

MCX MCU portfolio builds on the legacy of LPC and Kinetis, **bringing together the best of the both worlds**

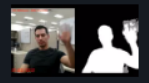
Industry platform **that scales from low-cost MCUs to wireless MCUs, to Linux based microprocessors**

GoPoint is part of the FRDM Ecosystem for i.MX Application Processors

Showcases a variety of use cases and applications, such as AI, video and graphics packed with Linux, Debian and Yocto BSP

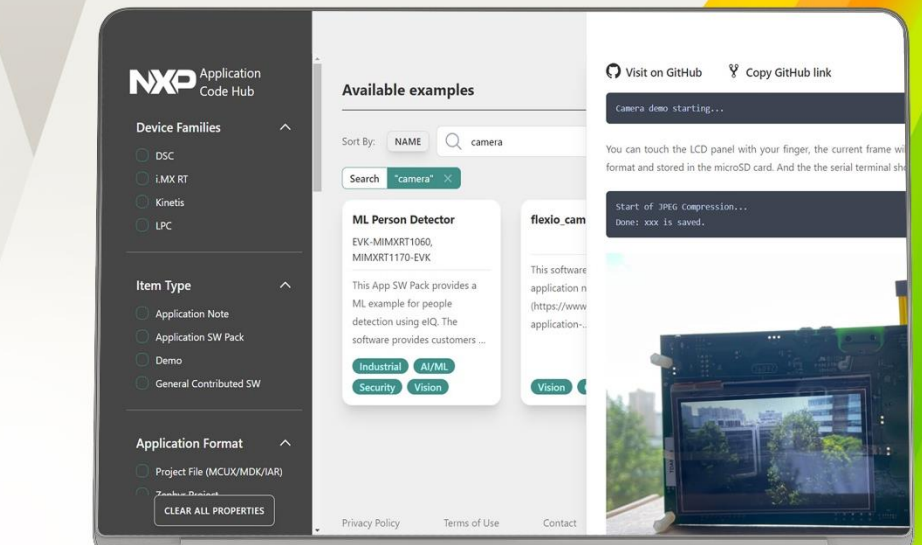
Launch your demos in seconds:

- Simple Out of Box Experience
- Simple GUI
- Continuing Support

Name	Description	8MM	8QM	8MP	93	95
 Image classification	Image classification example using NNStreamer. Image classification is an ML task that attempts to comprehend an entire image as a whole. The goal is to classify the image by assigning it to a specific label. Typically, it refers to images in which only one object appears and is analyzed.	✅	✅	✅	✅	✅
 Object detection	Object detection example using NNStreamer. Object detection is the ML task that detects instances of objects of a certain class within an image. A bounding box and a class label are found for each detected object.	✅	✅	✅	✅	✅
 Pose estimation	Pose estimation example using NNStreamer. The goal of pose estimation is to detect the position and orientation of a person or object. In human pose estimation, this is usually done with specific keypoints such as hands, head, legs, etc.		✅	✅		✅
 ML Benchmark	This tool allows to easily compare the performance of TensorFlow Lite models running on CPU (Cortex-A) and NPU. The tool works on i.MX93 and i.MX8M Plus.			✅	✅	
 ML Gateway	ML Gateway easily configures the i.MX8M Plus and i.MX93 EVKs as machine learning accelerator servers and allows resource-constrained MPU systems (clients) without NPUs to connect and run ML inference. This is currently enabled for i.MX8M Mini on the client side.	✅		✅	✅	
 Selfie Segmenter	Selfie Segmenter showcases the ML capabilities of i.MX8M Plus and i.MX93 by using the NPU to accelerate an instance segmentation model. This model lets you segment the portrait of a person and can be used to replace or modify the background of an image.			✅	✅	

See all the available demos at GoPoint

The power of the FRDM Ecosystem together with NXP's Rich Application Examples



[Application Code Hub](#) Features & Benefits

- Intuitive way to find source code for demos
- Interactive dashboard to filter and find resources.

One place to find everything:

- **SDK** software examples
- Demos and Application Examples from **Git repository**
- **GoPoint** demos from Git repository
- **Application notes** with its source code

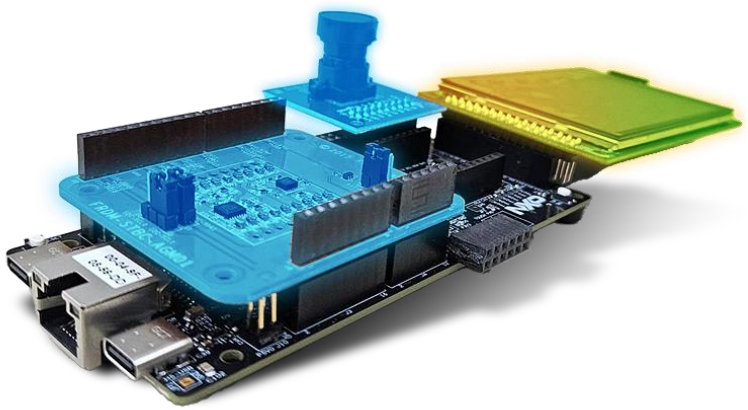
Use it together with [Expansion Board Hub](#) & create cool solution-demos!

Find expansion boards that are software supported from different boards by SDK, Git or Application Code Hub examples and use it together with different FRDM and Evaluation boards.

Fast Evaluation with FRDM Platform

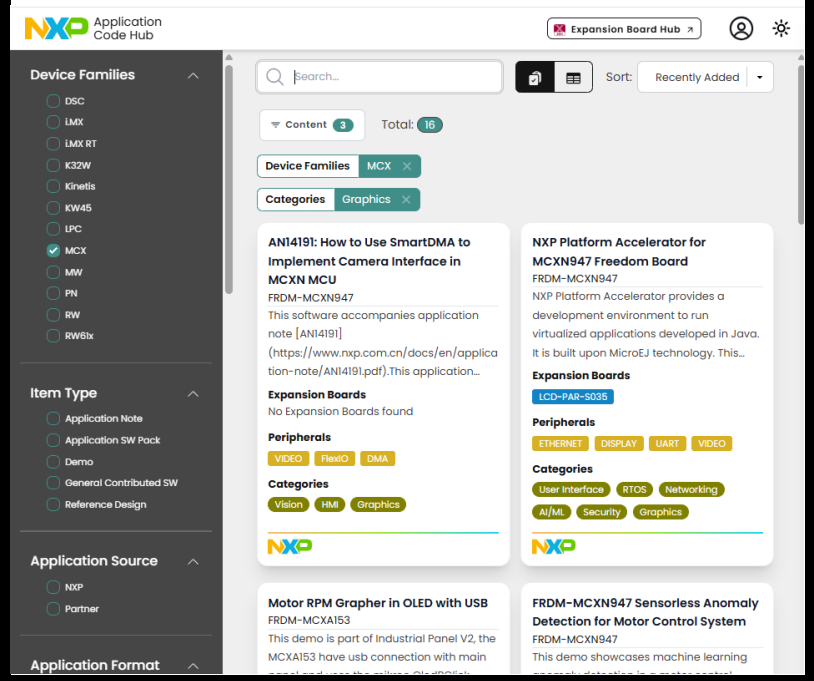
Explore a variety of **Open-source application code examples** based on the **FRDM platform**, designed for use cases like motor control, sensing, networking, graphics, machine learning, and more.

Application Code Examples are **developed by NXP experts and partners**, offering robust support within a rich software ecosystem



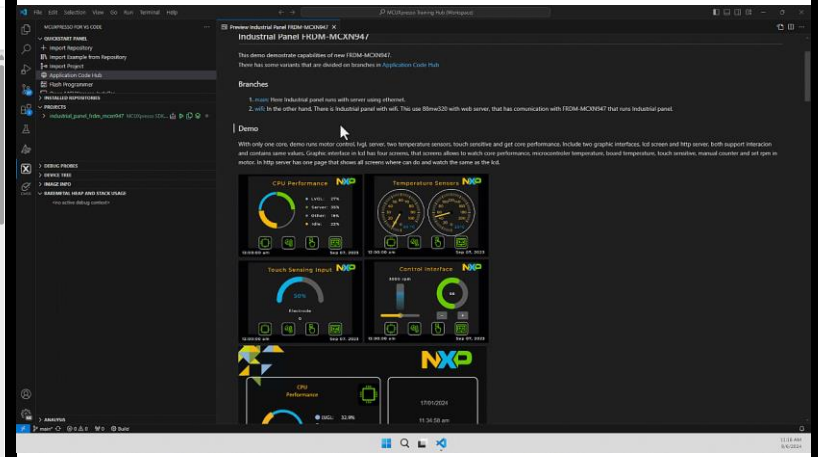
Run Application Code Hub demos from **the GitHub repository**, complete with a **step-by-step guide** for easy set up.

Use the interactive dashboard to filter and access available application code efficiently.



Application Code Hub enable streamlined setup and development for embedded applications.

Learn more how use ACH



Highlighted code examples in Application Code Hub

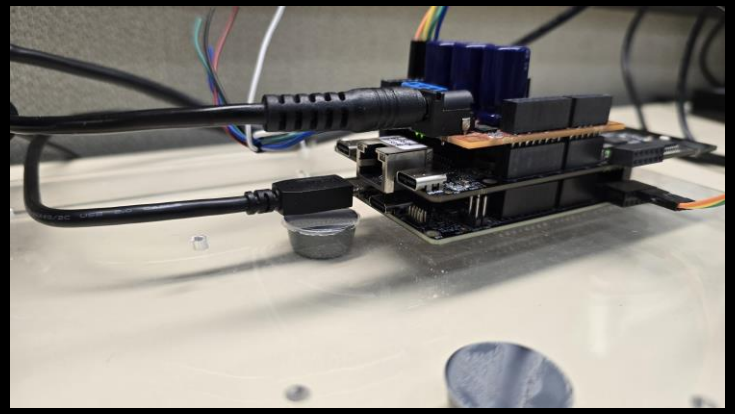
Motor Control of PMSM with NCP in FRDM-MCXA156

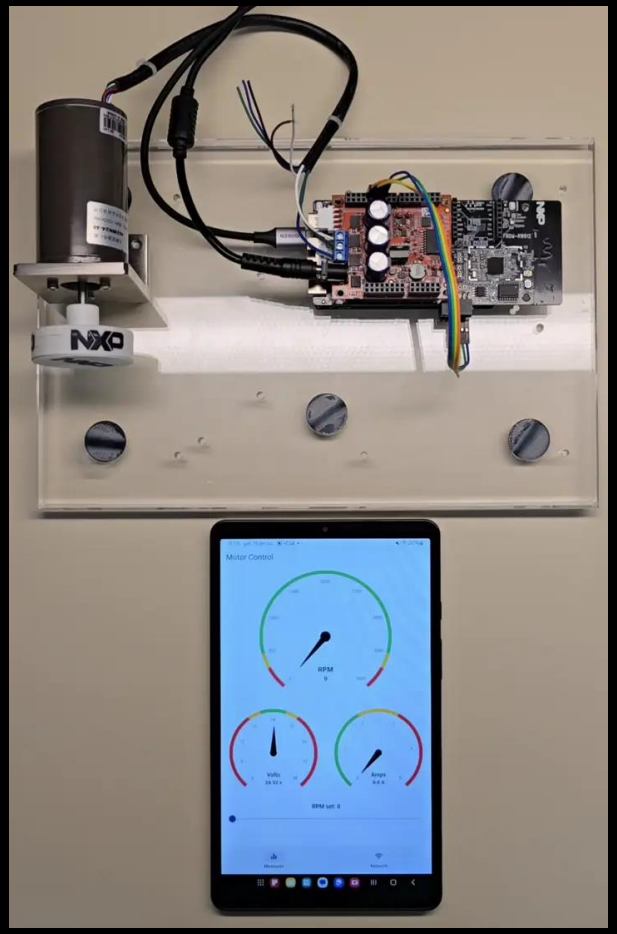
This application highlights the use of the Network Coprocessor (NCP) architecture, where the FRDM-RW612 functions as the dedicated connectivity coprocessor.

In this setup, the FRDM-RW612 creates the wireless access point and manages socket-based communication with the Android application. Meanwhile, the FRDM-MCXA156 operates as the main controller responsible for executing the PMSM motor-control algorithm.

Boards used:

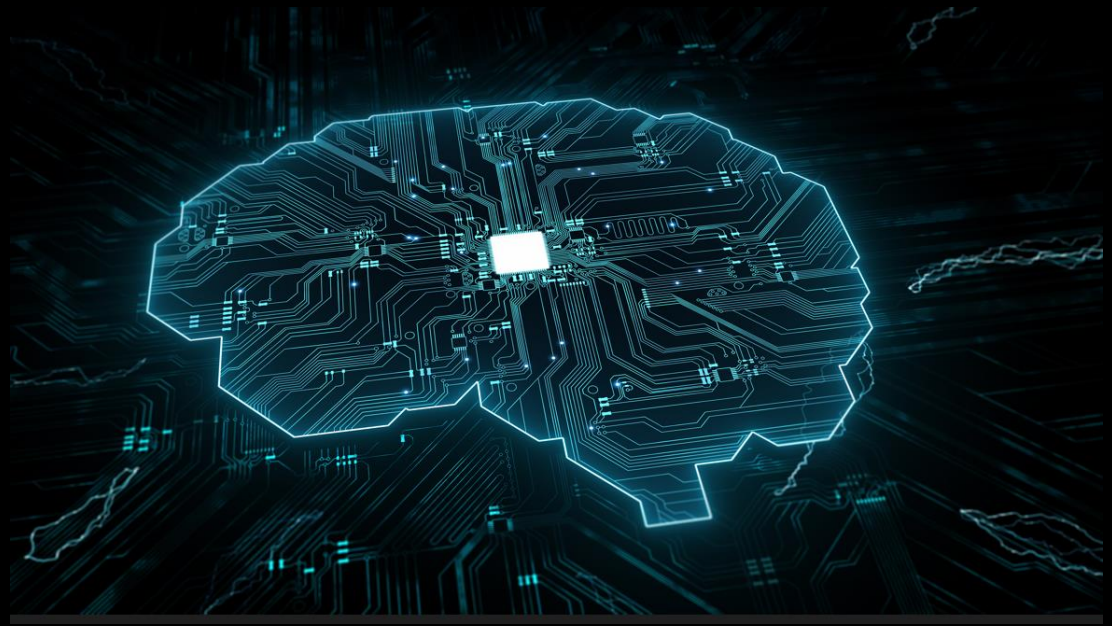
- FRDM-MCXA156
- FRDM-RW612
- FRDM-MC-LVPMSM
- FRDM-MC-LVMTR

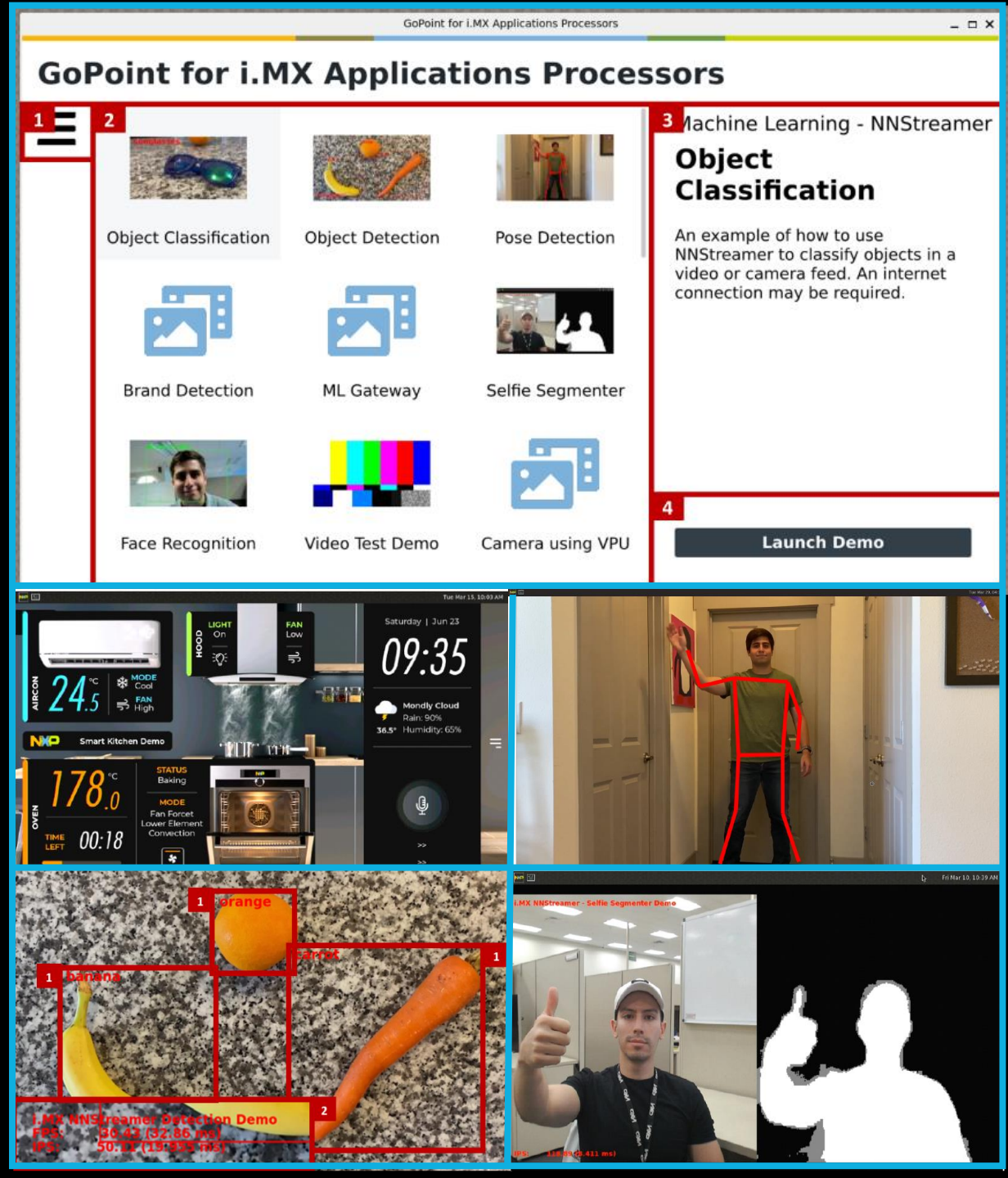




Gopoint

- Example applications built into Yocto releases
- User friendly interface to quickly and easily access demos
- Long term support





GoPoint = Fast Evaluation with FRDM Platform for i.MX Processors

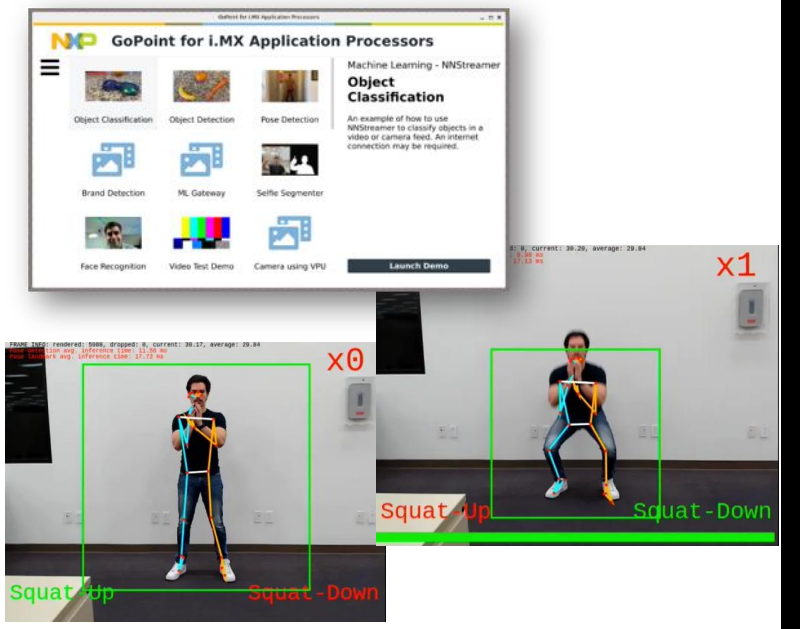
Coming soon

OOB Demo experience

Showcases a variety of use cases and applications with GoPoint **packed with Linux, Debian and Yocto BSP**

Launch your demos in seconds:

- Part of Out of Box Experience
- Simple GUI
- Continuing Support



What can I find in Go-Point?

You'll find ready-to-use demos for Vision, TSN, graphics, voice, and more—plus a variety of supported open-source examples to explore at official GitHub.

GoPoint for i.MX Applications Processors - Example Applications Package

This repository contains example applications supported by GoPoint for i.MX Applications Processors.

Machine Learning

Name	Description	8MM	8QM	8MP	93	95
Image classification	Image classification example using NNStreamer. Image classification is an ML task that attempts to comprehend an entire image as a whole. The goal is to classify the image by assigning it to a specific label. Typically, it refers to images in which only one object appears and is analyzed.	✅	✅	✅	✅	✅
Object detection	Object detection example using NNStreamer. Object detection is the ML task that detects instances of objects of a certain class within an image. A bounding box and a class label are found for each detected object.	✅	✅	✅	✅	✅
Pose estimation	Pose estimation example using NNStreamer. The goal of pose estimation is to detect the position and orientation of a person or object. In human pose estimation, this is usually done with specific keypoints such as hands, head, legs, etc.		✅	✅		✅
ML Benchmark	This tool allows to easily compare the performance of TensorFlow Lite models running on CPU (Cortex-A) and NPU. The tool works on i.MX93 and i.MX8M Plus.			✅	✅	

FRDM Development Boards Supported:

- FRDM-IMX93
- FRDM-IMX8MPLUS
- FRDM-IMX95

Easy set up

FRDM Development Platform offers options to start evaluating SoC with minimal and easy set-up.



Highlighted GoPoint Demos

Drive Monitor System

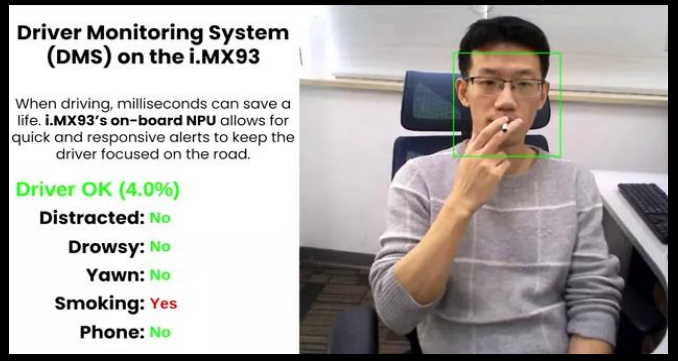
This demo showcases a real-time Driver Monitor System running on i.MX8M Plus and i.MX93 Evaluation Kits. It leverages the Neural Processing Unit (NPU) to accelerate ML inference for detecting driver fatigue and distraction, using a camera-based vision pipeline.

Key Features:

- **Real-time facial analysis** for detecting drowsiness and inattention
- **NPU acceleration** for low-latency inference

Benefits:

- Enhances vehicle safety by proactively identifying risky driver behavior
- Demonstrates edge AI performance on embedded platforms
- Reduces CPU load by offloading inference to NPU



i.MX Smart Kitchen

This demo simulates a voice-controlled smart kitchen. It showcases real-time voice command recognition to operate virtual appliances such as a hood, oven, and air conditioner.

Key Features

- Wake word activation: “Hey Hood”, “Hey Oven”, “Hey Aircon”
- Command recognition for specific functions (e.g., “Fan High”, “Open Door”)
- Touchscreen/mouse input as an alternative control
- NXP’s GUI Guider for advanced graphics and animations

Benefits

- Demonstrates real-time voice interaction on embedded systems
- Highlights low-latency, offline voice recognition
- Showcases intuitive user interface and multi-modal control

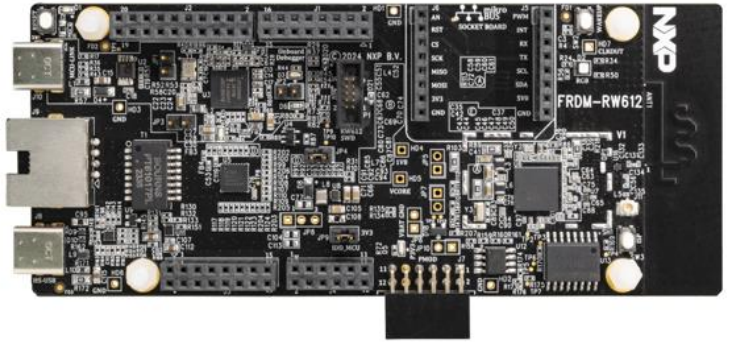




Get Started with NXP's FRDM Development Platforms for Wireless Connectivity!

STEP 1

Prototype with a FRDM board

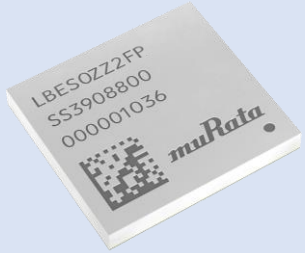


Wi Fi
Bluetooth
zigbee
THREAD
matter

STEP 2

Integrate wireless into your design using a module





NXP partners with leading module manufacturers to offer certified wireless solutions

STEP 3

Finalize your product



Get to market faster with NXP's Wireless Community Support

Get Started with NXP's FRDM Development Platforms for Wi-Fi Connectivity!

FRDM-RW612



- **Compact and scalable** development board
- Powered by **RW612** Wi-Fi 6 i.MX RT MCU
- **External flash** and on-board **MCU-Link Debugger**

Get Started

FRDM-IW416-AW-AM510



- **Wi-Fi 4 + Bluetooth** FRDM Expansion Board
- Compatible with **FRDM-MCXN9x**
- Based on the **Azurewave AW-AM510 Wi-Fi Module**

Get Started

FRDM-IMX91



- **Cost optimized** entry-level IoT platform
- Powered by **i.MX91** processor and **IW610**, IoT optimized Wi-Fi 6 Tri-radio
- **Linux** support and **eMMC** pre-programmed with SW BSP

Get Started

FRDM-IMX93



- **Low cost** and **compact** development board
- Powered by **i.MX93** processor and **IW612**, high performance Wi-Fi 6 Tri-radio
- Supports **GoPoint** for i.MX application processors

Get Started

i.MX Application Processors FRDM Development Platform

- Low-cost Linux Platform for Industrial and IoT applications.
- Rich set of interfaces, PMIC and Tri-Radio Wi-Fi 6, BLE and 802.15.4 module.
- Linux Debian and Yocto BSP support.
- FRDM page with step-by step guide.

Entry-Level i.MX91

Arm® Cortex®-A55
Parallel interface for camera and display

FRDM-IMX91S
- **Very tiny Linux Yocto** , with file system optimized.
- Debian support though SD Card.

FRDM-IMX91
FRDM Platform with **full Linux support**.

Mainstream i.MX93

2x Arm Cortex®-A55 + Cortex®-M33 and NPU.

Pin-to-pin compatibility with FRDM-IMX91 silicon package

Multimedia interfaces: MIPI CSI/DSI and LVDS to HDMI.

Advanced i.MX8MP

4x Arm Cortex®-A53 + Cortex®-M7, ML Accel

ML, advance multimedia and **industrial networking**

Multimedia interfaces: MIPI-CSI/DSI, HDMI, LVDS

Advanced i.MX95

4x Arm Cortex®-A53 + Cortex®-M7, ML Accel

Gen AI, Robotics, advanced multimedia and networking

M.2 connector for **Ara240 Discrete NPU**

Multimedia interfaces: MIPI-CSI/DSI, LVDS to HDMI, LVDS

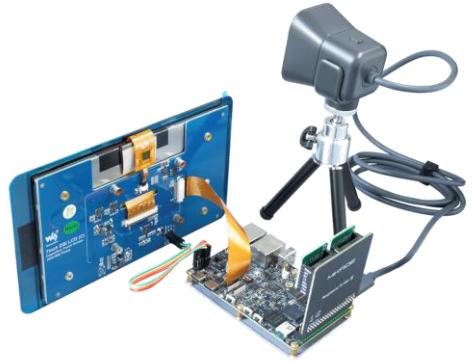
GoPoint , intuitive interface that provides easy access to application-specific demos

Starting at $50





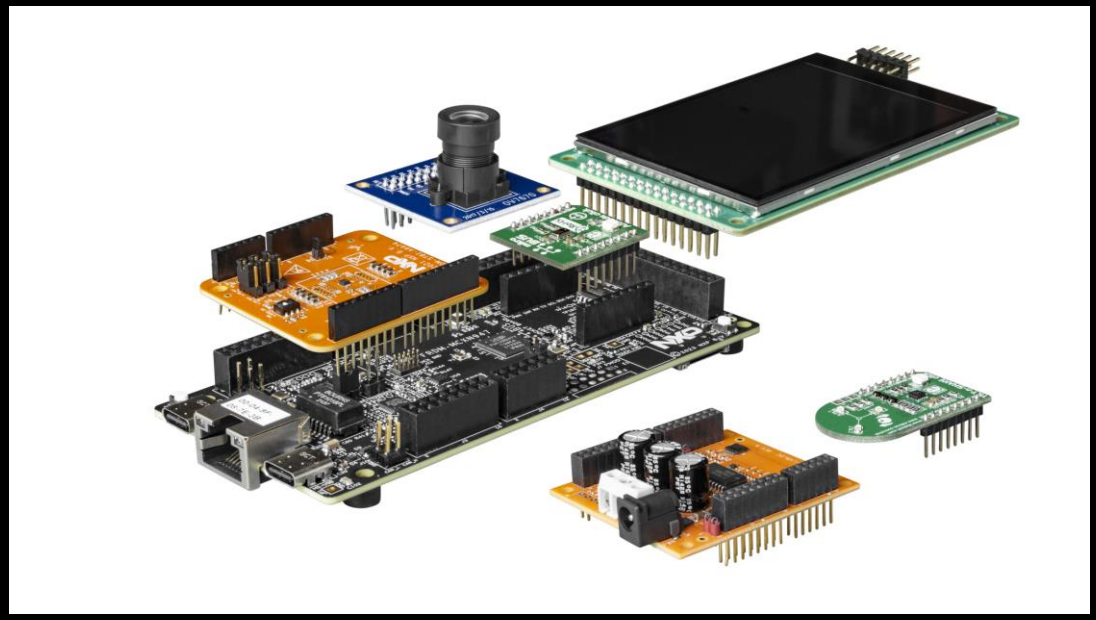




MCU FRDM Portfolio
Expansion boards with software drivers & examples

43 Expansion boards just for MCX FRDM Development Boards

- Displays
- Connectivity
- Motor Control
- Security
- Sensing
- RTC
- LED Drivers
- Touch
- LIN Bridge
- Adapters
- Relays
- And more …



All the expansion boards can be found at
mcuxpresso.nxp.com/eb-hub

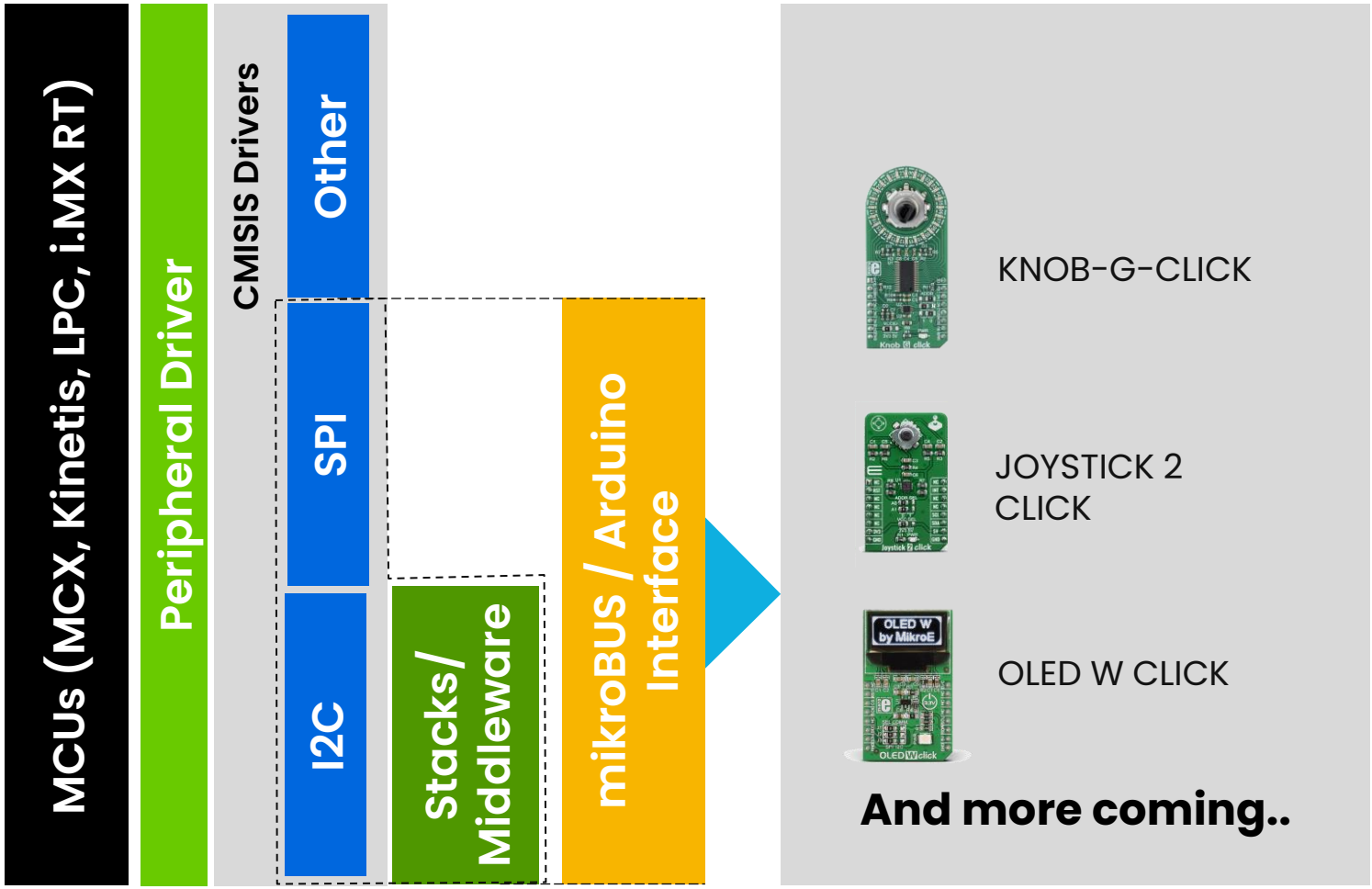
FRDM-IMX93 our first FRDM board with an MPU
Expansion boards

LCD Panel		i.MX95	i.MX 8MPLUS	i.MX93	i.MX 91	i.MX91S
DPI Kippah from Adafruit (ID:2453)	5.0" Display - 800x480 with Touchscreen			X		
TM050RDH03-HC	5.0" Display - 800x480			X	X	X
7inch DSI LCD ©	7inch Capacitive Touch IPS Display for Raspberry Pi, with Protection Case, 1024×600, DSI Interface	X	X	X		
DY1212W-4856	DY1212W-4856 TFT LCD panel with LVDS interface		X			
Camera						
RPI-CAM-MIPI	IAS camera to RPi camera adapter	X	x	X		
MY-CAM012B	Parallel Camera Module					X
EXPI-OS08A20	HR 4K CMOS Camera Module	X	x			
Audio						
MX93AUD-HAT	Audio expansion board with multi-channel codec, S/PDIF and microphone interfaces	X		X	X	X
8MIC-RPI-MX8	8-microphone array proto board for voice enablement	X	X	X	X	X
M.2 connector						
Ara240 16GB M.2 Module	Discrete NPU (Key M)	x	X			
Wireless module	WIFI/BT module	IW612	IW612	IW612	IW610	IW610G

See more compatible boards at mcuxpresso.nxp.com/eb-hub

Reuse software with CMSIS Driver Implementation

- CMSIS Drivers let you reuse software between devices



- Available ISSDK (drivers, algorithms and examples) and ACH examples based on CMSIS drivers
- Easy SW reuse across MCU portfolio
- Fast setup with a variety of expansion boards
- OLED
- Rotary
- Joystick
- And more…

Growing Community

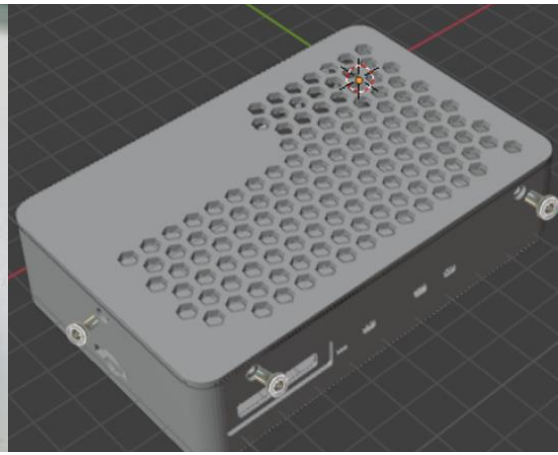
- [FRDM Training Hub] as part of developer journey
  - Highlighted content
    - [Getting Started with the FRDM Development Platform: A Complete Beginner's Guide]
    - [FRDM Trainings and resources]

- Makers content with design files to build and contribute to FRDM environment











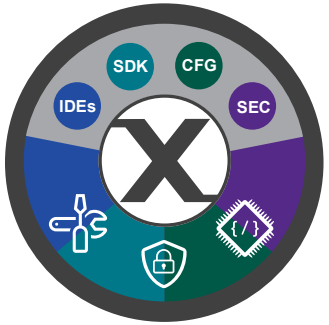


FRDM Growing Developer Experience

Enabling FRDM for real fast prototyping

- **Scalable** FRDM board portfolio from low-cost MCUs to Linux based i.MX Application Processors
- **Low-Cost** Portfolio (MCU $10-$35, MPU $50-$150)
- **22 boards** in production + **2026 FRDM** board Roadmap
- **150** SW Application Examples **ready to use** just for **FRDM**
- **43** entries in **Expansion Board Hub** just for **FRDM**
- Unified developer experience through all FRDM boards
- **Partners contributions** in Application Code Hub
- New Zephyr content in Application Code Hub
- **Zephyr** support for all MCX FRDM & **FRDM-IMX91 and FRDM-IMX93**
- 43 Articles in FRDM Training Hub (FRDM knowledge base)

FRDM is not just a board,
FRDM is a rich ecosystem



GitHub

EBH

SDK

Visual Studio Code

IDE

ACH

Flexible & Rapid Development Platform

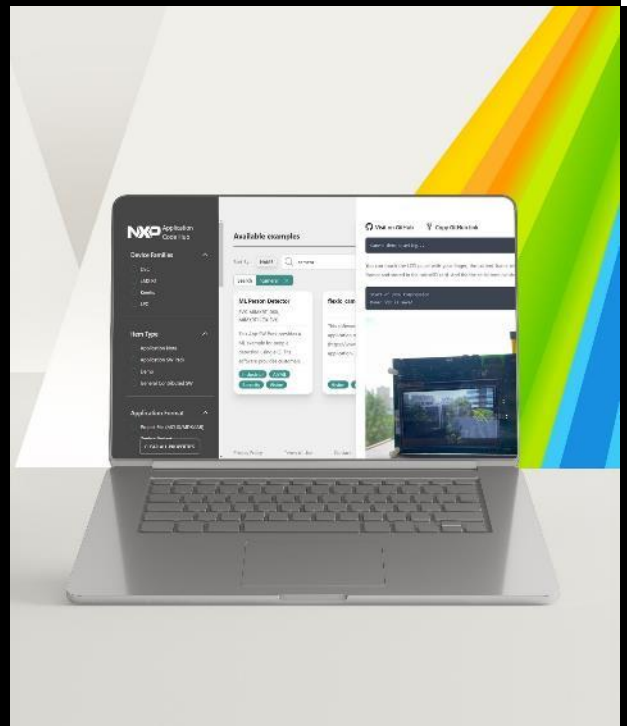
Explore FRDM Development Platforms at nxp.com/FRDM



Keep learning though our different channels:

- FRDM Blogs
- FRDM Training Hub
- FRDM Demo videos

Find it.
Get it.
Use it.



Design without bounds

